

Exercise 3: Machine Learning Basics

Task 1:

Do you agree with the following statement?

"Unsupervised learning is not suitable for classification problems, but supervised learning and reinforcement learning are."

Give reasons for your answer.

Task 2:

For each of the machine learning tasks below, provide a suitable metric from the lecture and explain your choice:

- Spam emails: Classify emails as either spam or not spam. False negatives (spam emails that slip through) are acceptable, but false positives (legitimate emails marked as spam) are annoying.
- Medical Diagnosis: Predict whether a patient has a rare disease based on test results. Missing a positive case is much worse than a false alarm.
- Rental prices: Estimate the monthly rent for a given apartment based on features such as size, location, and amenities. A large error is not particularly worse than multiple small errors.
- Weather: Predict tomorrow's average temperature in °C using weather data. Both large and small errors matter, but large deviations should be penalized more.
- Credit card fraud: Detect fraudulent transactions among many legitimate ones. Positive cases are rare, and you care about identifying them reliably, but you still don't want to spend too much time and money on investigating false alarms.

Task 3:

Explain what the training and test datasets are and what they are used for.

Discuss why a machine learning model should not be evaluated on the same data it was trained.

Task 4:

Check out the Jupyter Notebook *ml_basics.ipynb* from Moodle. This notebook will revisit the *titanic.csv* dataset and go over:

- Processing training and testing data
- Your first "predictive" model
- Evaluating machine learning models